Team Members:

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1. Project Title: RUSS: RAG-Based tUtor for Specific Subjects

2. Short Description of Goals and Limitations

- RUSS is a localized, Retrieval-Augmented Generation (RAG) system designed to act as a “Private NotebookLM” for cadets at the Air Defense College.
- **The goal** is to ingest complex technical manuals (PDFs) into a secure vector store and allow students to query them using natural language.
- **Limitation:** The prototype is restricted to text and tables within engineering textbooks (no image analysis) and is strictly air-gapped (no internet access allowed).

3. What Will You Achieve?

- **Learn:** We will master the end to end RAG Pipeline Implementation using Langchain. Specifically, we will implement Hybrid Search (combining sparse keyword search with dense semantic search) to ensure high accuracy retrieval for military terms. We will also learn to deploy Local LLMs (Llama 3 or Mistral) using Ollama.
- **Produce:** A functional local hosted web interface (Streamlit) where a user can upload a curriculum PDF and receive citation backed answers from a Llama3 model running entirely offline.
- **Prior Knowledge:** Proficiency in Python, understanding of API structures, and basic knowledge of how LLMs process text.

4. Why Do You Want To Do This Project?

We want to solve the "Pain of Secrecy" in military education. Cadets face a dilemma: they need modern AI assistance to understand complex engineering concepts, but security protocols strictly ban cloud-based tools like ChatGPT. RUSS bridges this gap by proving that we can build a secure, intelligent tutor that operates inside an isolated network, ensuring data sovereignty while modernizing the learning experience.

5. Final Deliverable & Users

- **Deliverable:**
  A Python-based application containing the RAG engine trained on specific subjects (ex: Essential MATLAB for Engineers and Scientists) and a Streamlit frontend. The key feature will be the "Source Verification Footer," which displays the exact page number and raw text snippet for every AI-generated answer to prevent hallucinations.
- **Target Users:**
  Cadets and instructors at the Egyptian Air Defense College (and eventually broader Military Academy branches).

6. Project Boundaries (30-Hour Scope)

- **Included:**
  - **Smart Parsing:** Processing engineering PDFs (using Marker or PyMuPDF) to preserve equations and tables.
  - **Storage:** Storing embeddings locally using ChromaDB.
  - **Retrieval:** Implementing a Hybrid Retriever (BM25 + Vector Similarity) to handle specific military jargon.
  - **Generation:** Using Llama 3 (8B-Instruct) via Ollama as the reasoning engine.
  - **Grounding:** Strict prompt engineering to ensure the model acts as a military instructor and provides citations.
  - **Interface:** A simple Streamlit chat interface.
- **Excluded:**
  - Multi-modal RAG (analyzing images/diagrams from the books).
  - Fine-tuning the model (we are using RAG only)
  - Integration with official academy grading systems or student records.
  - Chat history persistence (sessions will be temporary).